

Q.bloxx A111

Measurement Module for IEPE Sensors and Voltages

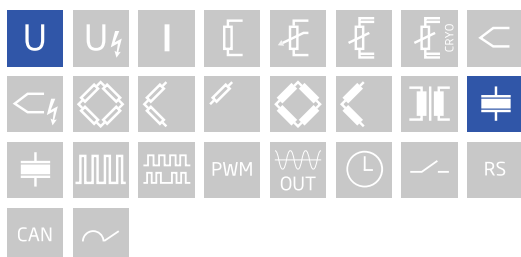
Q.bloxx is the ideal DAQ solution for widely distributed installations, electrical panels, and environmental enclosures. Q.bloxx measurement modules provide integrated signal conditioning and arithmetic functions, packaged in modular, DIN Rail mountable enclosures that easily snap together for quick system expansion. Flexibility in distribution allows for highly synchronized data that is less prone to noise due to shorter sensor cable runs to the actual point of measurement.

- RS 485 fieldbus interface up to 24 Mbps: LocalBus up to 115.2 kbps: Modbus-RTU, ASCII
- Connectable to any Controller, e.g. Q.station, Q.gate or Q.pac
- Electromagnetic Compatibility according to EN61000-4 and EN55011
- Power supply 10 ... 30 VDC
- DIN rail mounting (EN60715)

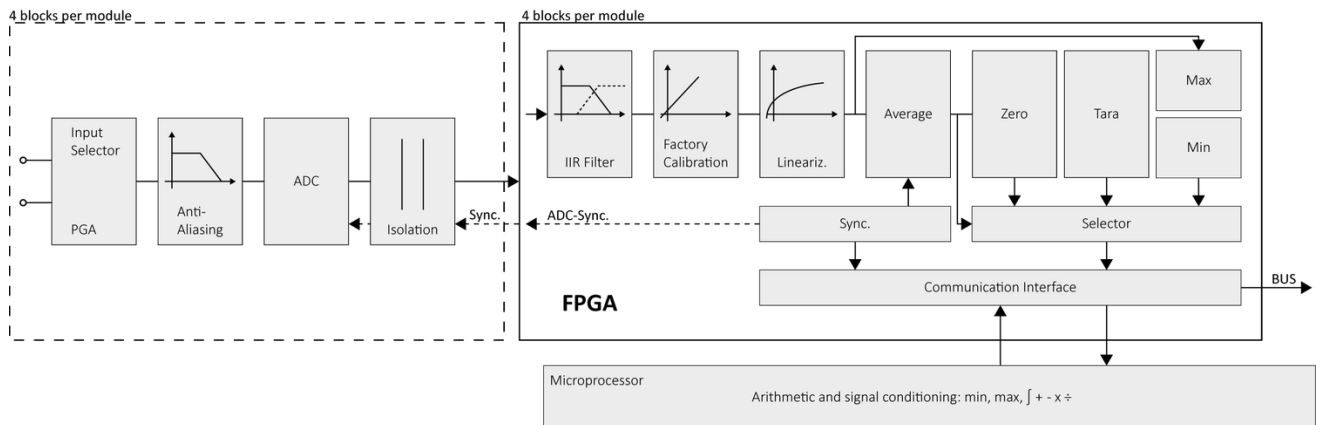


Key Features

- 4 galvanic isolated analog input channels
IEPE sensors, voltage
- Configurable input ranges
 ± 100 mV, ± 1 VDC, ± 10 VDC
- High-accuracy digitization
24-bit ADC, 100 kHz sample rate per channel
- Signal conditioning
16 virtual channels, linearization, digital filter, average, scaling, min/max storage, RMS, arithmetic, alarm
- Galvanic isolation
Channel to channel, channel to power supply, and bank



Block diagram

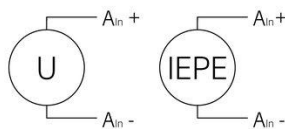


Technical Data

Terminal assignment 10pole screw

1	-	
2	A _{in} 1+	
3	-	
4	A _{in} 1-	
5	-	
6	-	
7	A _{in} 2+	
8	-	
9	A _{in} 2-	
10	-	

1	-	
2	A _{in} 3+	
3	-	
4	A _{in} 3-	
5	-	
6	-	
7	A _{in} 4+	
8	-	
9	A _{in} 4-	
10	-	



Analog Input

Channels	4
Accuracy	0.01 % typical
	0.025 % in controlled environment ¹
	0.05 % in industrial area ²
Linearity error	0.01 % typical full-scale
Repeatability	0.003 % typical (within 24 hrs)
Input impedance	>10 MΩ (unless otherwise stated)
Isolation voltage	500 VDC channels, to power supply, channel to bus ³
Overvoltage protection	±30 V
Max. Common-mode voltage (CMV)	250 VDC

¹ according to EN 61326 2006: appendix B

² according to EN 61326 2006: appendix A

³ noise pulses up to 1000 VDC, continuous up to 250 VDC

Measurement Mode Voltage

Input range	Margin of error	Resolution	Input impedance
±100 mV	±20 µV	12 nV	>1 MΩ
±1 V	±200 µV	120 nV	>1 MΩ
±10 V	±2 mV	1.2 µV	>1 MΩ
Long-term stability (range ±1 V)	<20 µV / 24 hrs		
Temperature drift (range ±1 V)	<50 µV / 10 K Offset drift		
Signal-to-noise ratio	>90 dB at 1 kHz		
Dynamic range	109 dB @ ±10 V		
Input impedance	1.2 MΩ 330 pF		

Measurement Mode IEPE

Input range	Margin of error	Resolution	Input impedance
±1 V	±1 mV	120 nV	>1 MΩ
±10 V	±10 mV	1.2 µV	>1 MΩ
Sensor excitation	4 mA ±10% constant current		
Compliance voltage	24 VDC ±10%		
Input frequency range	0.5 Hz to 20 kHz		
Temperature drift (range ±1 V)	<50 µV / 10 K Offset drift		
	<0.025 % / 10 K Gain drift		

Analog/Digital Conversion

Resolution	24-bit
Sample rate	100 kHz per channel
Modulation method	sigma-delta
Anti-aliasing filter	20 kHz, 3rd order
Digital filters	Infinite impulse response (IIR), low-pass, high-pass, Butterworth or Bessel (2nd, 4th, 6th or 8th order), frequency range 0.1 Hz to 10 kHz (adjustable via software)
Averaging	configurable or automatic according to the selected data rate

Communication Interface

Protocols	proprietary Localbus (115200 bps to 24 Mbps, latency <100 ns) ASCII (19200 bps to 115200 bps) Modbus RTU Profibus-DP (19200 bps to 12 Mbps) (special Firmware required)
Data format	8E1
Electrical standard	ANSI/TIA/EIA-485-A, 2-wire

Power Supply

Input voltage	10 to 30 VDC, overvoltage and overcurrent protection
Power consumption	2.5 W (approx.)
Input voltage influence	<0.001 % / V

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Measurement Module for IEPE Sensors and Voltages

Environmental

Operating temperature	-20°C to +60°C
Storage temperature	-40°C to +85°C
Relative humidity	5 % to 95 % at 50°C, non-condensing

Remarks

Are subject to a warm-up period of at least 45 minutes

Specifications subject to change without notice

Mechanical information

Material	Aluminum and ABS
Measurements (W x H x D)	27 x 120 x 105 mm
Weight	approx. 200 g

Ordering Information

Article number	696538
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